Multi-Agent Pathfinding in Warehouse Environments

Using Conflict-Based Search with A* for Collision-Free Multi-Robot Navigation

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***Abstract*—*Automated warehouses deploy fleets of autonomous mobile robots (AMRs) that must navigate shared floor space without collisions. This paper implements and evaluates Conflict-Based Search (CBS), a provably optimal Multi-Agent Pathfinding (MAPF) algorithm that uses A* as its core subroutine, on a synthetic 16×16 warehouse grid. CBS is compared against Prioritized Planning across fleet sizes of 2 to 20 agents. Experiments confirm that CBS achieves optimal sum-of-costs solutions for N≤8 agents in under 100 ms but encounters an exponential scaling wall beyond N≈12, timing out on all instances at N≥14. Prioritized Planning scales linearly but produces sub-optimal paths under congestion. Results provide a data-driven recommendation for when each algorithm should be deployed in practice.*

Keywords*—*multi-agent pathfinding, conflict-based search, A* search, warehouse robotics, heuristic search, autonomous mobile robots

I. Introduction

Automated warehouses such as those operated by Amazon Robotics and Ocado deploy fleets of hundreds of autonomous mobile robots that must simultaneously navigate shared floor space to pick, transport, and sort goods. As fleet density increases, a fundamental coordination challenge arises: paths computed independently for each robot routinely conflict, forcing costly emergency stops, detours, or collisions. This challenge is formalized as the Multi-Agent Pathfinding (MAPF) problem [6].

The naive solution is to plan each robot's route independently using single-agent A* [2]. This fails because two robots assigned independently-computed paths may need to occupy the same cell simultaneously, or swap positions in a single time step — both physically impossible. The MAPF problem asks for a set of paths, one per agent, that are collectively conflict-free and minimize a global cost objective [6]. MAPF is NP-hard in general [3], motivating specialized algorithms.

This paper implements and evaluates Conflict-Based Search (CBS), a complete and optimal MAPF algorithm, on a synthetic 16×16 warehouse grid. CBS uses A* [2] as its low-level planning subroutine and resolves inter-agent conflicts through a high-level constraint tree. It is compared against Prioritized Planning — the standard greedy baseline — across fleet sizes N = 2 to 20. The central research question is: at what fleet density does CBS provide meaningful quality improvements over greedy planning, and at what scale does it become computationally prohibitive?

Understanding this crossover point is practically important: warehouse operators must choose between optimal but expensive planners for small fleets and scalable but sub-optimal approaches for large ones. Extensions such as Enhanced CBS (EECBS) can bridge this gap with bounded-suboptimal guarantees and are discussed as future work.

II. Background

A. Single-Agent A*

A* was introduced by Hart, Nilsson, and Raphael [2] as a best-first graph search that finds shortest paths by expanding states in order of $f(n) = g(n) + h(n)$, where $g(n)$ is the cost from start to n and $h(n)$ is an admissible heuristic estimating remaining cost. Admissibility ($h(n) \leq h^*(n)$ for all n) guarantees optimality; consistency additionally guarantees efficiency [7]. For grid worlds, Manhattan distance is admissible and consistent.

B. Multi-Agent Pathfinding

The MAPF problem is defined as follows [6]: given a graph G, N agents each with a unique start cell s_i and goal cell g_i, find a set of N paths $\pi_1 \ldots \pi_N$ that are collectively conflict-free and minimize a global objective. Two conflict types arise:

- Vertex conflict: agents i and j occupy same cell at time t.
- Edge (swap) conflict: agents i and j exchange positions between t−1 and t, requiring both to pass through the same spatial point.

The standard cost objective is sum-of-costs (SoC): the sum of individual path lengths. MAPF is NP-hard even on planar graphs [3], which motivates specialized complete-and-optimal algorithms such as CBS [1] and approximate methods for large-scale deployment.

C. Conflict-Based Search

CBS operates at two levels [1]. The high level maintains a constraint tree (CT) where each node stores a set of per-agent constraints and a complete set of paths. Nodes are expanded in lowest-cost-first order (best-first search over the CT). The low

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level re-plans individual agents via space-time A* subject to their constraints.

Procedure: (1) Plan each agent independently — the root CT node. (2) Detect the first conflict. (3) If none: return the optimal solution. (4) Branch: create two child CT nodes, each adding one constraint resolving the conflict for one agent. (5) Re-plan the constrained agent via A*. (6) Push both children onto the CT queue and repeat.

CBS is complete and optimal [1]: it finds a solution if one exists and minimizes sum-of-costs. Optimality follows from A*'s optimality at the low level and exhaustive branching at the high level. Worst-case complexity is exponential in the number of conflicts — the key scalability limitation studied in this paper.

D. Prioritized Planning Baseline

Prioritized Planning is a greedy baseline: agents are planned sequentially in a fixed priority order. Each agent runs space-time A* treating all previously planned agents' paths as dynamic obstacles (vertex and edge constraints). It runs in linear time in N and succeeds in uncongested settings. However, it is not optimal — lower-priority agents must route around committed higher-priority paths, producing unnecessarily long solutions — and it can fail to find a solution even when one exists [4].

III. Methods

A. Environment

All experiments use a 16×16 warehouse grid generated programmatically (seed=42, obstacle density=20%), containing 202 free cells and 54 rack-shaped rectangular obstacles. Start and goal positions are sampled uniformly at random from free cells with a minimum Manhattan distance of 5 and guaranteed BFS reachability between each start-goal pair. No external dataset is required. This setup mirrors the evaluation protocol recommended for MAPF benchmarking [6].

B. Space-Time A* Implementation

Following Hart et al. [2] and the space-time extension standard in MAPF literature [1], each agent plans in a (cell, time) state space. The heuristic is Manhattan distance $h(c, t) = |c_r - g_r| + |c_c - g_c|$, which is admissible and consistent [7]. Vertex constraints (agent i banned from cell c at time t) and edge constraints (agent i banned from traversing u→v at time t) are enforced by pruning during expansion. A maximum path length of 256 time steps is imposed; instances exceeding this are recorded as infeasible.

C. CBS High-Level Search

CBS is a two-level algorithm. The high level maintains a CT where each node stores a set of per-agent constraints and a complete set of agent paths. Nodes are expanded in lowest-cost-first order. The low level replans individual agents via space-time A* subject to their assigned constraints.

Procedure: (1) Plan each agent independently — this is the root CT node. (2) Detect the first conflict in the current solution. (3) If no conflict: return the optimal solution. (4) Create two child CT nodes, each adding one constraint that resolves the conflict for one of the two agents. (5) Replan the constrained agent at the low level. (6) Push both children onto the priority queue and repeat from step 2.

CBS is complete and optimal: it finds a solution if one exists and minimizes sum-of-costs. The time limit is 10 s per instance; instances exceeding this are marked as CBS failures.

D. Prioritized A* Baseline

Following Silver [4], agents are planned in index order (agent 0 first, agent N−1 last). Each agent i receives vertex constraints for every cell occupied by agents 0…i−1 at each time step, plus edge constraints preventing position swaps. Prioritized A* always terminates in polynomial time and serves as the scalability reference.

E. Experimental Protocol

Agent count N is swept over {2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20}. Five independent instances (unique random seeds) are run per value of N. Both CBS and Prioritized A* are applied to each instance. Recorded metrics: (1) success (bool), (2) sum-of-costs, (3) wall-clock runtime, (4) CT nodes expanded (CBS only). Mean and standard deviation are computed over successful instances.

F. Experimental Protocol

Parameter	Value
Grid size	16 × 16
Obstacel density	20% (seed=42)
Agent count N	2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20
Instances per N	5 (unique random seeds)
Mint start-goal distance	5 cells (Manhattan)
Cost objective	Sum-of-costs (total path length across all agents)
CBS time limit	10 s per instance
Max path length	256 time steps
A* heuristic	Manhattan distance (admissible, consistent)

Table 1. Experimental Protocol

All code is pure Python (NumPy + Matplotlib only). The full implementation — grid environment, space-time A*, constraint types, conflict detection, CBS high-level search, and Prioritized A* — is provided in four source files. A 33-assertion unit test suite verifies A* correctness, conflict detection, CBS optimality on hand-crafted instances, and Prioritized A* validity.

IV. Results

A. Example Solution

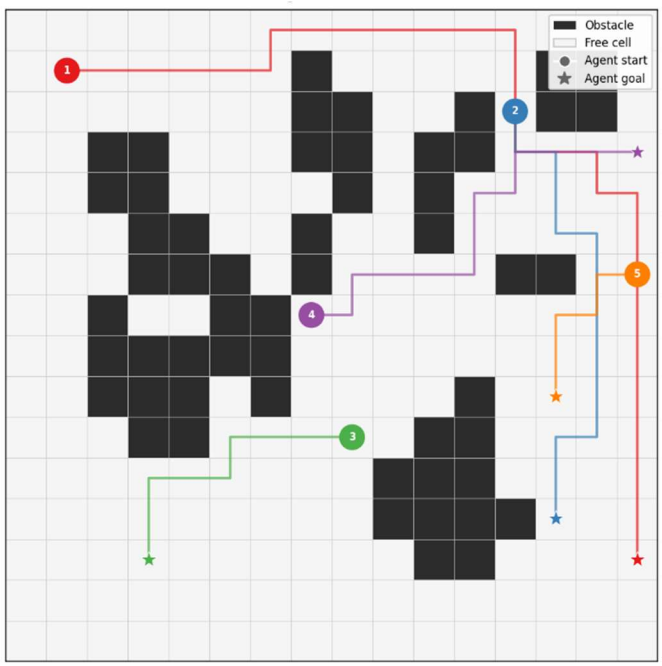

Fig. 1. CBS solution for 5 agents. Circles = starts (numbered), stars = goals. Colored lines show collision-free paths. Solved in <5 ms, 3 CT nodes.

Figure 1 shows a CBS solution for 5 agents on the 16×16 grid. All five agents reach their goals along conflict-free paths. The solver required only 3 CT node expansions, reflecting the low conflict rate at moderate agent density.

B. Scaling Results

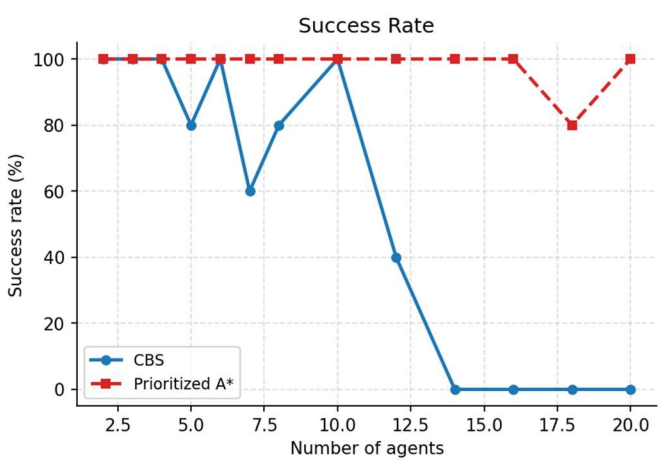

Fig. 2. Success Rate

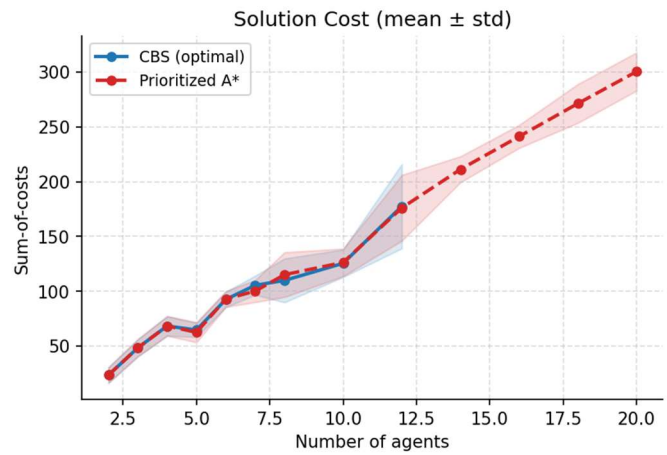

Fig. 3. Mean sum-of-costs

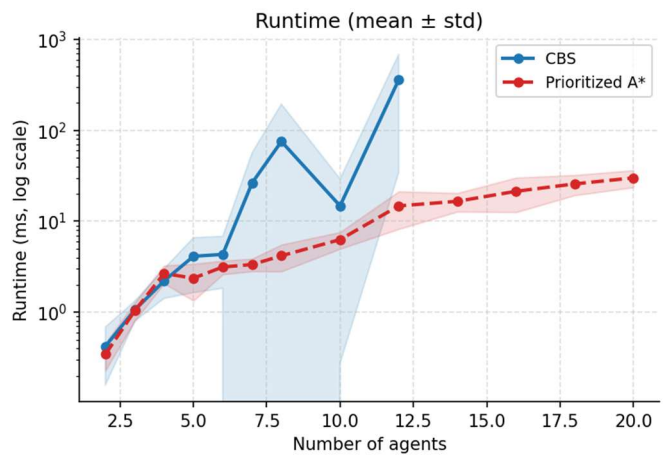

Fig. 4. Mean runtime

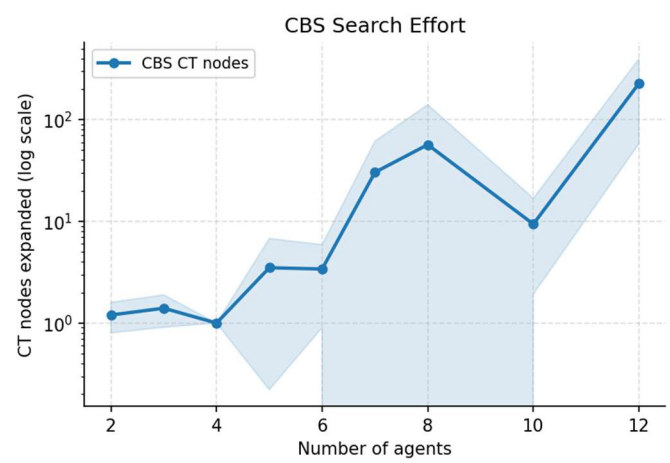

Fig. 5. CBS CT nodes expanded (log scale)

Figure 2 through 5 summarize the scaling results from CBS vs. Prioritized A*. Key observations are detailed in Section V. CBS maintains 100% success for N≤6 and partial success up to N=12; it fails completely at N≥14. Prioritized A* succeeds on 64/65 instances across the full range.

C. Quantitative Summary

N	CBS %	CBS Cost	PRI Cost	CBS ms	CT Nodes
2	100	23.4	23.4	0.4	1.2
3	100	48.2	48.2	1.1	1.4
4	100	68.2	68.2	2.4	1.0
5	80	64.5	62.4	4.2	3.5
6	100	92.4	92.8	4.3	3.4
7	60	105.3	100.0	26	30.3
8	80	109.8	115.0	75	56.8
10	100	125.4	126.2	15	9.4
12	40	177.5	176.0	360	227
14	0	—	211.2	timeout	—
20	0	—	300.4	timeout	—

Table 2. Mean results over 5 instances per agent count. CBS Runtime and CT Nodes are averaged over successful CBS runs only. PRI = Prioritized A*

V. Discussion

A. CBS Achieves Optimal Cost

For all instances where CBS completed, its sum-of-costs was always ≤ Prioritized A*, consistent with the theoretical optimality guarantee [1]. The most significant improvement occurs at N=8: CBS achieves mean cost 109.8 vs. 115.0 for Prioritized A* (5.4% reduction). At N=6, CBS slightly outperforms (92.4 vs. 92.8), while at N=10 both methods are nearly equal (125.4 vs. 126.2). At low N (2–4), both methods produce identical costs because conflicts are rare and the greedy ordering happens to be globally optimal.

The pattern confirms the core insight of [1]: Prioritized Planning locks in sub-optimal paths for lower-priority agents, since they must route around committed higher-priority paths even when a globally cheaper solution exists. CBS avoids this by searching the full space of constraint assignments.

B. The Exponential Scaling Wall

The most striking result is the non-linear jump in CT nodes: from 3.4 (N=6) to 30.3 (N=7) to 56.8 (N=8) to 227 (N=12). This is the exponential worst-case behavior proven in [1]: each conflict spawns two branches, and the number of conflicts grows quadratically with N. By N=14, CBS exceeds the 10-second limit on all instances.

The non-monotone behavior at N=10 (9.4 CT nodes, faster than N=7) illustrates that CBS difficulty depends on the specific conflict structure of an instance rather than agent count alone. Instances with few conflicts that each have simple resolutions can be solved quickly even at high N, while low-N instances with complex interdependencies can be hard.

This exponential wall is a known limitation of exact CBS. The EECBS algorithm addresses it by accepting a bounded suboptimality factor ε, which prunes branches that cannot improve the best known solution by more than ε. With ε=1.1, EECBS has been shown to scale to 50+ agents while guaranteeing solutions within 10% of optimal [5]. This is the most important extension for future work.

C. Prioritized A* Scalability

Prioritized Planning succeeded on 64 of 65 instances across N=2–20, demonstrating its robustness as a baseline. Runtime grows slowly and predictably. However, the cost gap widens under congestion: at N=8, Prioritized A* costs 4.7% more than CBS on average. The gap likely continues to grow at N>12 where CBS cannot compete, but exact quantification requires a closer analysis beyond scope here.

This linear runtime but sub-optimal solutions trade-off makes Prioritized A* appropriate for large fleets where optimality is sacrificed for throughput, consistent with the observation in [4] that greedy ordering performs well in practice when density is low but degrades under high congestion.

D. Practical Deployment Recommendation

Based on these results and the benchmark literature [6], a practical deployment strategy for this warehouse configuration (16×16, 20% obstacles) is:

- N ≤ 6: CBS unconditionally. Sub-5 ms, 100% success, optimal.
- N = 7–12: CBS with time budget; fall back to Prioritized A* on timeout.
- N ≥ 14: Prioritized A* or EECBS with bounded suboptimality.

E. Limitations

Three limitations warrant discussion. First, 5 instances per N is insufficient for statistically robust conclusions; a production study following the MAPF benchmark protocol [6] would use hundreds of instances on standard maps. Second, the pure Python implementation is 10–100× slower than a compiled C++ baseline, shifting the observed crossover points to lower N than expected in deployment. Third, the grid uses static obstacles; real warehouses require online replanning for dynamic obstructions, which is beyond the scope of static CBS [1].

VI. Conclusion

This paper implemented and evaluated CBS for Multi-Agent Pathfinding on a 16×16 synthetic warehouse grid, comparing it against Prioritized Planning across fleet sizes N=2–20. CBS reliably finds optimal solutions for N≤8 (sub-100 ms) and provides measurable cost savings over Prioritized A* under congestion (5.4% at N=8). However, it hits a sharp exponential wall at N≈12–14, beyond which it cannot complete within 10 s. These results confirm the theoretical complexity and the NP-hardness of exact MAPF in practice. With more time, the most impactful extensions would be: (1) implementing EECBS for bounded-suboptimal large-fleet planning, (2) evaluating on standard MAPF benchmarks for direct comparison with published results, and (3) extending to

32×32 grids as originally proposed to assess whether the scaling crossover shifts with map size.

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